Case Study: E-Commerce Sales & Customer Insights

Business Problem

An e-commerce company is experiencing:

- Fluctuating sales performance
- High return rates in certain categories
- Unclear customer purchasing patterns

Dataset Overview

Your dataset includes:

- **OrderID** – Unique order identifier
- **CustomerID** – Unique customer ID
- **OrderDate** – Date of purchase
- **Category** – Product category (Electronics, Clothing, etc.)
- **Price & Quantity** – Unit price and number of items
- **TotalAmount** – Before discount
- **Discount & FinalAmount** – Pricing after discount
- **PaymentMethod** – Payment type used
- **City** – Customer location
- **DeliveryDays** – Delivery time
- **Returned** – 1 = Returned, 0 = Not returned

BUSINESS REQUIEMENT :-

1. Sales Performance Analysis

- Monthly revenue trends
- Top-performing categories
- City-wise revenue contribution

2. Customer Behavior

- Repeat vs new customers
- Average order value (AOV)
- Preferred payment methods

3. Returns Analysis

- Return rate by category
- Correlation: delivery time vs returns
- High-risk products

4. Discount Impact

- Do higher discounts increase sales?
- Impact on profitability

5. Operational Insights

- Delivery performance by city
- Relationship between delivery time and customer satisfaction (returns proxy)

STAKEHOLDER MANAGEMENT :-

Stakeholder	Role	INTERSET LEVEL	Influence Level
Business Manager	Decision-making, strategy	High	High
Marketing Team	Campaign performance	High	Medium
Operations Team	Delivery & logistics	Medium	Medium
Finance Team	Revenue & profitability	High	High
Customers	End users	High	Low

BUSINESS INSIGHTS :-

1. Sales Performance

The company generated total revenue of ₹651K, with noticeable fluctuations across months, indicating seasonal trends. Electronics and Home categories emerged as top contributors to revenue.

---

2. Customer Behavior

The Average Order Value (AOV) is ₹651, reflecting moderate customer spending per transaction. A significant proportion of customers prefer online payment methods, highlighting a shift toward digital transactions.

---

3. Returns Analysis

The overall return rate stands at 51%, which is considerably high and may negatively impact profitability. Certain product categories exhibit higher return rates, suggesting potential issues in product quality, description accuracy, or customer expectations.

---

4. Operational Efficiency

Delivery performance varies across cities, with longer delivery times potentially leading to increased return rates. Improving logistics and delivery timelines can enhance customer satisfaction.

---

5. Discount Strategy

While discounts are frequently applied, their impact on increasing sales appears limited. Excessive discounting may be affecting profit margins, indicating the need for a more optimized pricing strategy